Fields, Affine Fibers, and the Linear Algebra of Forgetting Directions

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A vector space owes much of its simplicity to division by nonzero scalars. Over $\mathbb{C}$ this is already visible in the identity

$$z^{-1} = \frac{\overline{z}}{|z|^2} \qquad (z \neq 0).$$

Yet the most useful distinction in early linear algebra concerns equations: a homogeneous equation defines a subspace, while an inhomogeneous equation defines either an empty set or a translate of a subspace. From that observation, quotient spaces and duality arise with surprisingly little extra machinery.[1]

§1.1 Complex multiplication is a real linear transformation

Write $z = a + bi$. Multiplication by z is an $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$, represented in the basis $1, i$ by

$$M_z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

The identity $M_z M_w = M_{zw}$ follows from complex multiplication itself, while

$$M_z^T M_z = (a^2 + b^2) I, \qquad \det M_z = a^2 + b^2.$$

For $z = re^{i\theta}$, the map is a rotation through θ followed by dilation by r. The algebraic inverse $1/z = \overline{z}/|z|^2$ and the matrix inverse $M_z^{-1} = M_{\overline{z}}/|z|^2$ are therefore two descriptions of the same operation.

For example, $\omega = e^{2\pi i/3}$ satisfies

$$1 + \omega + \omega^2 = 0, \qquad M_\omega^3 = I.$$

The nontrivial cube roots of unity are rotations of order three. They provide an early example of a representation of the cyclic group C_3 on a real vector space.

§1.2 What a linear equation actually determines

Let $T : V \to W$ be linear and consider $T(v) = b$. If v_0 is one solution, then

$$T(v) = b \quad \Longleftrightarrow \quad T(v - v_0) = 0,$$

so the full solution set is

$$\boxed{T^{-1}(b) = v_0 + \ker T.}$$

The fiber is a translate of the kernel. It is a vector subspace precisely when $b = 0$, provided that the equation is solvable.

[1] For the linear-algebraic setting, see Axler, *Linear Algebra Done Right*, and MIT 18.700 lecture readings: https://ocw.mit.edu/courses/18-700-linear-algebra-fall-2013/pages/readings/.

This viewpoint resolves a family of elementary tests at once. In F^4, the condition

$$x_3 - 5x_4 = b$$

defines a nonempty affine hyperplane; it is a linear subspace only when $b = 0$. In the vector space $C^1((0,3),\mathbb{R})$, the condition $f'(2) = b$ has the same structure: evaluation of the derivative at 2 is a linear functional, and its fibers are translates of its kernel.

The equations need not involve finite coordinates. If $T : C^1([0,1]) \to \mathbb{R}$ sends f to $\int_0^1 f(t)\,dt$, the functions with integral 1 form an affine fiber of the same kind.

§1.3 What Survives the Quotient

For a subspace $U \subseteq V$, two vectors are equivalent when $v - w \in U$. Their equivalence classes form the quotient V/U, with projection $q(v) = v + U$.

Proposition 1.3.1 (Factorization through a quotient)
If $T : V \to W$ is linear and $U \subseteq \ker T$, there is a unique linear map $\overline{T} : V/U \to W$ such that $T = \overline{T} \circ q$.

Proof. There is only one possible definition:

$$\overline{T}(v + U) = T(v).$$

If $v+U = w+U$, then $v-w \in U \subseteq \ker T$, hence $T(v) = T(w)$. The definition is therefore independent of the representative. Linearity and uniqueness follow from surjectivity of q. □

Taking $U = \ker T$ makes $\overline{T}$ injective, with image equal to $\operatorname{im} T$. Thus

$$\boxed{V/\ker T \cong \operatorname{im} T.}$$

The first isomorphism theorem says that a linear map records exactly the information left after the directions invisible to it have been identified. For finite-dimensional spaces, taking dimensions recovers rank–nullity.

§1.4 Annihilators and Quotients

The dual $V^* = \operatorname{Hom}_F(V, F)$ is the space of linear measurements on V. A functional $\lambda \in V^*$ descends to V/U precisely when it is constant on each coset, or equivalently when $\lambda(u) = 0$ for all $u \in U$.

Define the annihilator

$$U^\circ = \{\lambda \in V^* : \lambda|_U = 0\}.$$

Pullback along q gives the map

$$q^* : (V/U)^* \longrightarrow V^*, \qquad \varphi \longmapsto \varphi \circ q.$$

It is injective because q is surjective, and its image is exactly U°, by the quotient factorization just proved. Hence

$$\boxed{(V/U)^* \cong U^\circ.}$$

For $V = \mathbb{R}^3$ and $U = \operatorname{span}\{(1,1,0)\}$, a linear functional $\lambda(x,y,z) = ax + by + cz$ lies in U° exactly when $a + b = 0$. The two functionals $x - y$ and z are independent measurements on the quotient; they distinguish its cosets and form a basis of its dual.

This is a concrete reason to learn covectors. When one solves linear constraints, it is often easier to describe the surviving information as linear measurements than to choose arbitrary coordinates on the quotient.

§1.5 Why changing the scalars changes the theory

A vector space over a field has bases, and every linearly independent list extends to a basis. The mechanism is that every nonzero pivot coefficient can be inverted. For a module over a ring this may fail.

For example, $\mathbb{Z}/2\mathbb{Z}$ is a $\mathbb{Z}$-module but has no basis over $\mathbb{Z}$: if a putative free generator e existed, the relation $2e = 0$ would contradict uniqueness of integer coordinates. This is torsion. More subtly, even a torsion-free module need not be free over an arbitrary ring.

The distinction points directly toward integral matrix reduction. Over $\mathbb{Q}$, a nonzero integer pivot may be normalized to 1; over $\mathbb{Z}$, only ± 1 are units, and row and column operations preserve arithmetic divisibility information. Smith normal form is the corresponding normal form over $\mathbb{Z}$.

Quotients, annihilators, and affine fibers already make sense before any matrices are written down. Fields then guarantee that dimension and coordinates behave especially well, explaining both the efficiency and the limitations of elementary Gaussian elimination.